



Fig. 4: Seven types of tests to be done in the entire product life cycle

microcontroller (MCU) based, we can safely assume there is a need for design phase software/firmware development for testing the embedded system.

Board bring up code (BUC) and power-on self-test (POST)

These codes are dedicated software that will run in an MCU/processor-based board/system. A bring up code (BUC) is used to test a newly assembled board, section by section. BUC is typically interactive and user can feed parameters to vary the test data and exercise the newly developed board with various data.

Non-interactive version of the BUC is called POST or built-in self-test (BIST) and has a completely different use. POST/BIST is the first piece of software that will be executed in the manufactured product to diagnose the product every time the processor/controller boots up. It is shipped in every piece of product that is shipped out.

In the days of microprocessors (where all the memories and peripherals were separate devices), POST was critical as these components were external and many times from different vendors, increasing the need for critical testing. Best example of POST for microprocessors is the BIOS (basic input output system) in a PC.

The biggest challenge in the case of a microprocessor with the peripherals being outside the chip is that the software developer has to develop the code that uses internal processor resources to first test the memories and then execute the full POST/BIOS from the memory. With the advent of microcontrollers, most of these memories and peripherals are integrated into the same device, making the POST development simpler. However, testing the peripherals at the pin level is key to ensure PCBA has not been damaged and circuit interconnections are as per the design.

One big challenge is the testing of analogue-to-digital converter (ADC)

interfaces. ADC testing has two challenges—testing the basic functioning of the interface and the accuracy of the device. Most often, ADCs need a reference voltage and their functionality and accuracy is solely dependent on the reference voltage. Any change in the reference voltage can lead to system malfunction. Firmware developers who develop the BUC/POST code are required to have a complete understanding of the microprocessor/microcontroller and their peripherals, especially from the address map perspective, so that POST code is effective and compact.

As mentioned earlier, for bringing up a new PCBA, BUC is needed. Board BUC is basically the same POST code with interactive features. In fact, a smart firmware engineer can design a single firmware which can be either used as board BUC or POST, based on a dedicated signal from the board designer as an indication to the controller.

But we need to remember that a newly-assembled PCBA can have faulty interfaces, and this type of testing will be difficult if the device is defective or, in the case of microprocessor based systems, if there is a short in the PCBA forcing the POST code not to boot up. This situation calls for debugging with special tools, which is outside the scope of this article.

One of the common mistakes that hardware designers make is the lack of proper provisioning for programming the microcontrollers in the PCBA. There are no uniform programming techniques that device vendors follow. Most device vendors

use serial interfaces like SPI and other similar interfaces, customised to their development tool. There are also third-party tool vendors who offer universal programming, but they are expensive.

However, the global standard seems to provide a JTAG port. JTAG port is a serial interface specifically designed for testing the ICs and PCBAs in the manufacturing line in a short time. Most IC vendors have used the same JTAG for programming the devices too.

One of the key facts that designers should consider is the programming, whether bulk programming is feasible and whether the product is expected to be made in large volumes. Ninety five percent of the designers or companies have never factored this into their design and end up spending a lot of time and money in programming alone.

Remember, if your software is frozen, your microcontroller vendor can offer a pre-programmed device for a small fee, saving you lot of effort and investment. But this is feasible only when your product is stable and no changes are envisaged.

The job of the POST code in an embedded system is to do minimum diagnostics in the version that has to be shipped. POST tests the memories and tests essential interfaces that are needed before the main software boots up and takes over. When the board is assembled and the hardware engineers are testing, board BUC is the key tool that helps in bringing up the board, and with a little tweak this BUC can be converted into POST.

Most companies never factor in that a PCBA bring-up is a joint effort between the hardware and software designers. The tool that connects them is the programmers interface document (PID).

PID is a simple document which lists the addresses and the corresponding device registers that will be accessed and also gives out the details of every bit in the register/memory and what action they will perform. This is a document which, if prepared during early design phase, can save plenty of time later.

Now a days, many microcontroller vendors do offer graphical utilities to configure a particular microcontroller and the tool produces the initialisation code either in C or in Assembly, which

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